

## UNIT 3: DRIVES

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### A.C. MOTORS :

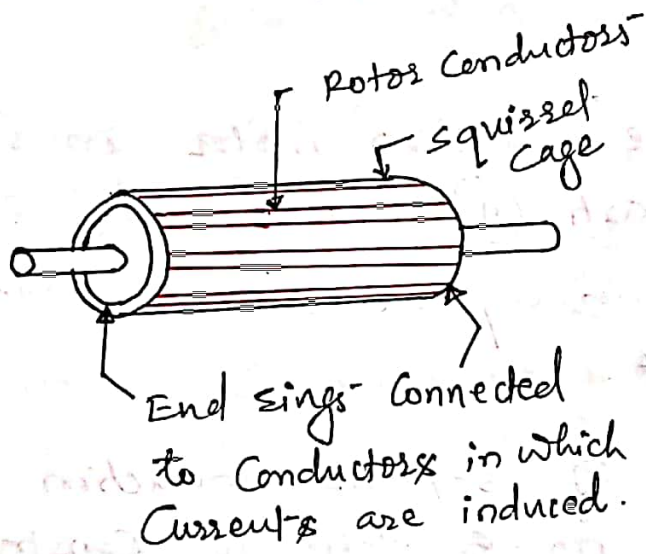
→ A.C. motors can be classified into two groups; 1) Single-phase and 2) polyphase, with each group being further subdivided into:

- 2.1) 3-phase induction motor
- 2.2) 3-phase synchronous motor.

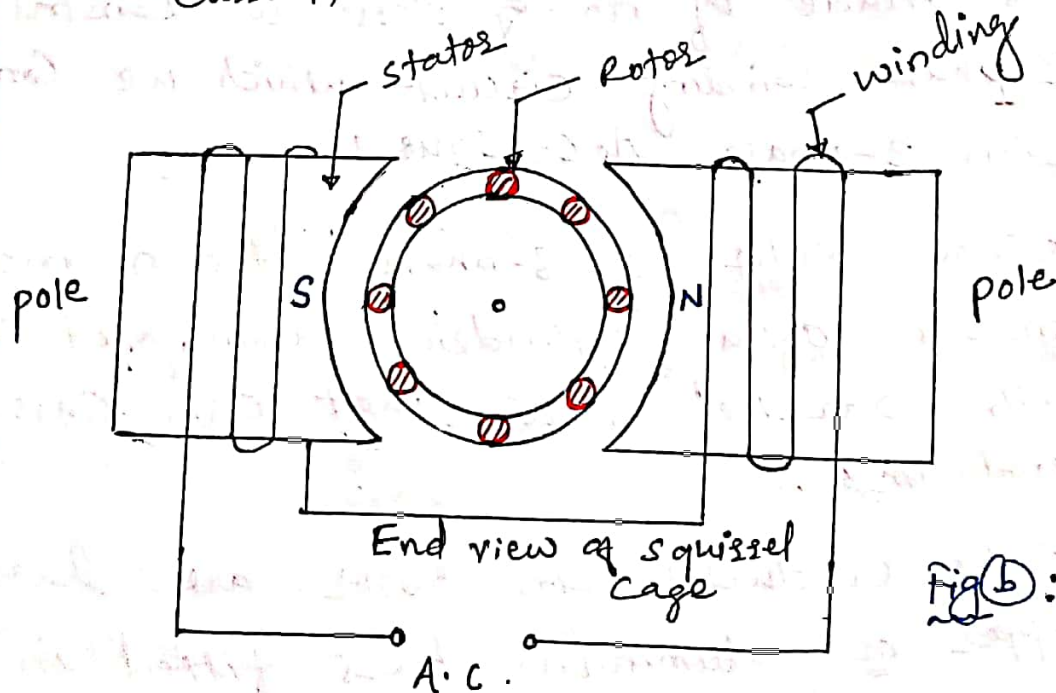
#### (1) Single-phase induction motor :

- Single phase induction motor consists of a squirrel cage-rotor.
- Copper or Aluminium bars are fitted into slots in endrings of squirrel cage rotor.
- There are no external electrical connections to the rotor.
- When an alternating current passes through the stator windings an alternating magnetic field is produced.
- As a result of electromagnetic induction, emfs are induced in the conductors of the rotor & current flows in the rotor.
- Initially, when the rotor is stationary & there is no net torque.

- The motor is not self-starting, an auxiliary starting winding to give the rotor an initial push is used.
- The rotor rotates at a speed determined by the frequency of A.C. applied to the stator.
- i.e. the speed of rotor depends on frequency of alternating current applied to the stator.



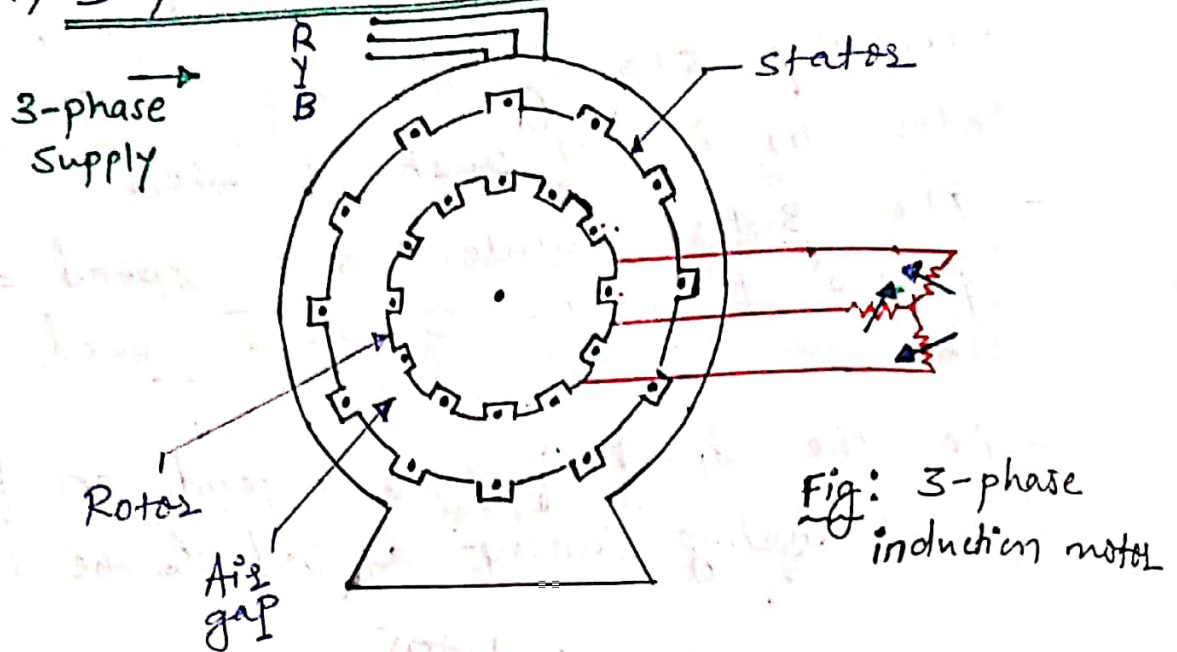
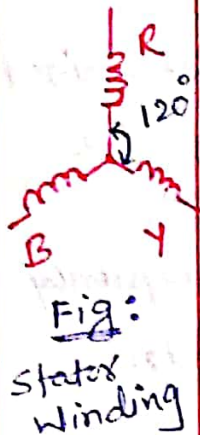
Fig(a): Squirrel Cage rotor



Fig(b):

Fig: Single-phase induction motor.

## (2) 3-phase induction motor:



- The 3-phase induction motor consists of a 2-main parts (a) stator and (b) rotor. Stator is stationary part while, Rotor is rotating part.
- The stator of 3-phase induction motor is made of no. of slots to construct a 3-phase winding circuit which we connect with 3-phase A.C. source.
- The Rotor of 3-phase induction motor consists of a cylindrical laminated core with parallel slots that can carry conductors.
- The conductors in rotor are heavy copper or aluminium bars fitted in each slot & fixed to end rings at the both ends.

→ The 3-phase induction motor is similar to the single-phase induction motor, but has a stator with three-windings located at  $120^\circ$  apart.

→ When we connect the stator to 3-phase A.C. source, it produces a rotating magnetic field. This magnetic field rotates at a synchronous speed.

→ The current will be induced in rotor conductors due to rotating magnetic field. When current starts flowing through conductors it reacts with the magnetic field. Hence, current carrying conductors experience a mechanical force due to which rotor rotates inside the stator.

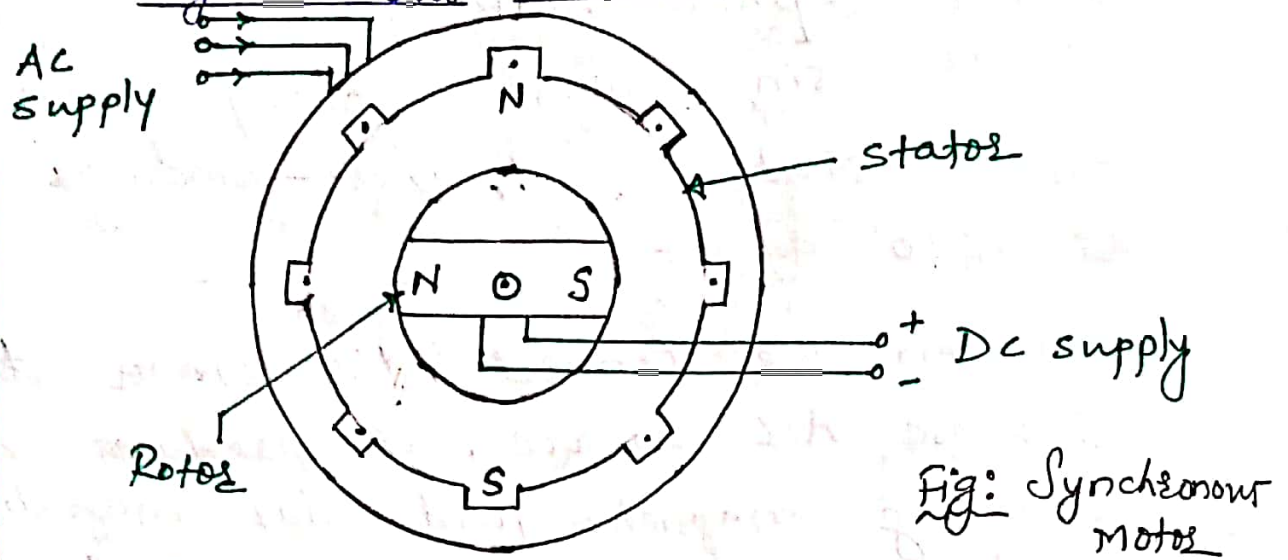
→ The rotation of the field is much smoother than with the single-phase motor.

→ The 3-phase induction motor is self-starting, which has a great advantage over the single-phase motor.

### Advantages :

- 1) self-starting.
- 2) Economical
- 3) Robust in construction.
- 4) Easier to maintain.

### (3) Synchronous Motors:



- The stator and rotor are the main parts of the synchronous motor. The stator is the stationary part & the rotor is moving i.e. rotating part of the m/c.
- The stator and rotor both are excited separately.
- When the 3-phase supply is given to the stator, the rotating magnetic field is developed between the stator & rotor gap. Because of rotating magnetic field south & North poles are developed on the stator.
- The rotor is excited by the D.C. supply. The D.C. supply induces the north & south poles on the rotor. As the D.C. supply remains constant, the flux induces on the rotor remains same. Thus, the flux has fixed polarity.

→ When the rotor & stator both have the same pole on the same side, they repel each other. If they have opposite poles, they attract each other.

→ The A.C. is sinusoidal. The polarity of the wave changes in every half cycle. The rotor attracts towards the pole of stator for the first half cycle & repulse for the second half cycle. Thus, the rotor becomes pulsated only at one place.

→ The north & south poles of the rotor and the stator interlock each other. Thus, the rotor starts rotating at the speed of the rotating magnetic field. And the motor runs at the synchronous speed (i.e. rotating magnetic field).

→ The speed of the motor can only be changed by changing the frequency of the supply.

### Advantages:

- 1) Speed is independent of load.
- 2) High efficiency
- 3) Maintains precise & constant speed

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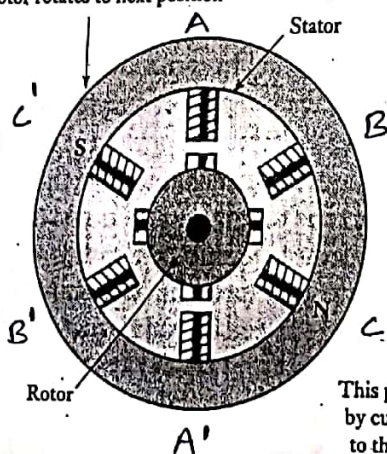
## STEPPER MOTORS :

Def<sup>n</sup>: The stepper motor is a device that produces rotations through equal angles, the so-called steps, By energizing each phase in sequence.

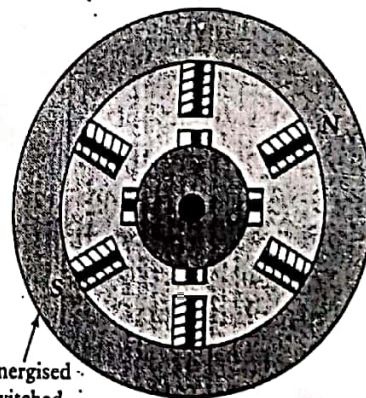
### 1). Variable reluctance stepper :

Figure 1.1 Variable reluctance stepper motor.

This pair of poles energised by current being switched to them and rotor rotates to next position



This pair of poles energised by current being switched to them to give next step



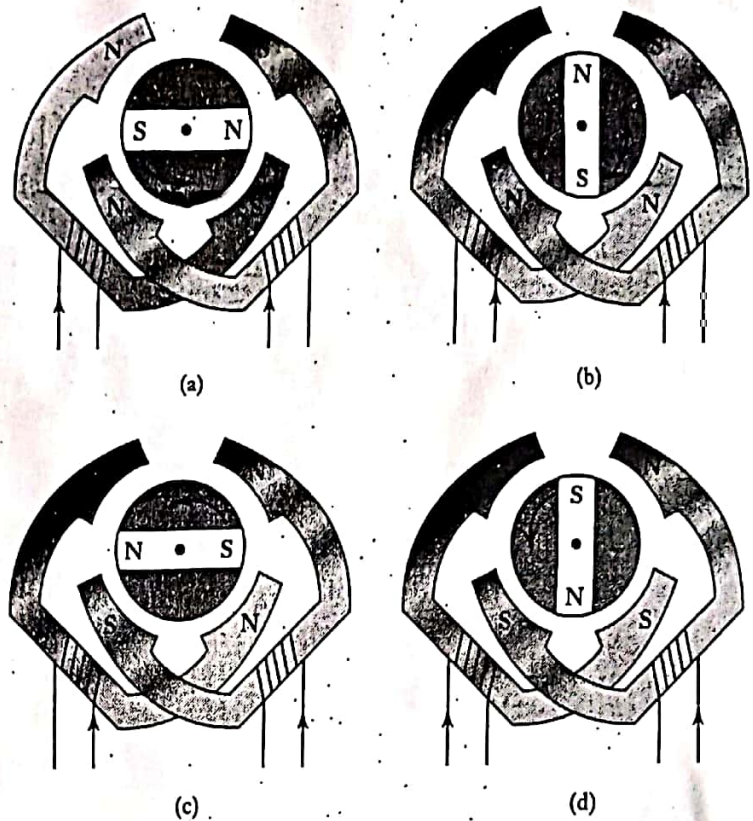
- Fig: shows the basic form of the variable reluctance stepper motor.
- The Rotor is made of soft-steel & is cylindrical containing 4-poles, fewer than the stator.
- When the current is switched to the opposite pair of windings,

a magnetic field is produced with lines of force.

- These lines of force pass from the stator poles to the rotor poles.
- Hence, the lines of force cause the rotor to move & rotate.
- Each winding (coil) is energized in sequence  $AA'$ ,  $BB'$  &  $CC'$  which cause the rotor to rotate in a series of discrete steps or angles.
- Increase of airgap between stator & rotor poles increases reluctance, while decrease of airgap between stator & rotor poles decreases the reluctance.
- This form of stepper motor generally gives step angles of  $7.5^\circ$  or  $15^\circ$ .
- The rotor will move until the rotor and stator poles are line-up (in a straight row). This is termed the position of minimum reluctance.

## 2) Permanent magnet stepper motor with 2-phase with $90^\circ$ steps :

Figure 2.2 Permanent magnet two-phase stepper motor with  $90^\circ$  steps. (a), (b), (c) and (d) show the positions of the magnet rotor as the coils are energised in different directions.



→ Fig: shows the basic form of the permanent magnet motor.

- The motor shown has a stator with four poles (i.e. 2-phase motor)

- Each pole is wound with a field winding, the coils on opposite pairs of poles being in series.

- The current is supplied from a d.c source to the windings through switches.

- When a current is supplied to the stator, the rotor which is a permanent magnet rotates & moves to lineup with the stator-poles.
- The rotor of the motor moves to the  $45^\circ$  position.
- If current is switched to another phase, the rotor will move a further  $45^\circ$  in order to line up again.
- Thus; by switching the currents through the coils in sequence of stator, the rotor rotates in  $90^\circ$  steps.
- With this type of motor, step angles are commonly  $1.8^\circ, 7.5^\circ, 15^\circ, 30^\circ, 34^\circ \approx 90^\circ$ .

## D.C Motors :

- A Direct Current (DC) motor is a type of electric machine that converts electrical energy into the Mechanical energy. D.C motors take electrical power through direct current, & convert this energy into mechanical rotation.

- D.C. motors are divided into two main groups (a) Brush & (b) Brushless type d.c motor.

### (a) Brush-type d.c motor :

- A brush-type d.c motor is essentially a coil of wire, which is free to rotate and so termed rotor, in the field of a permanent magnet or an electro-magnet.

- The magnet being termed as stator since it is stationary.

- When current is passed through the coil, the resulting forces acting on its sides at  $90^\circ$  angles to the field cause forces to act on those sides to give rotation.

- However, for the rotation to continue, when the coil passes through vertical position the current direction through coil has to be reversed & this is achieved by the use of brushes making contact with a split-ring commutator, the commutator rotating with the coil.

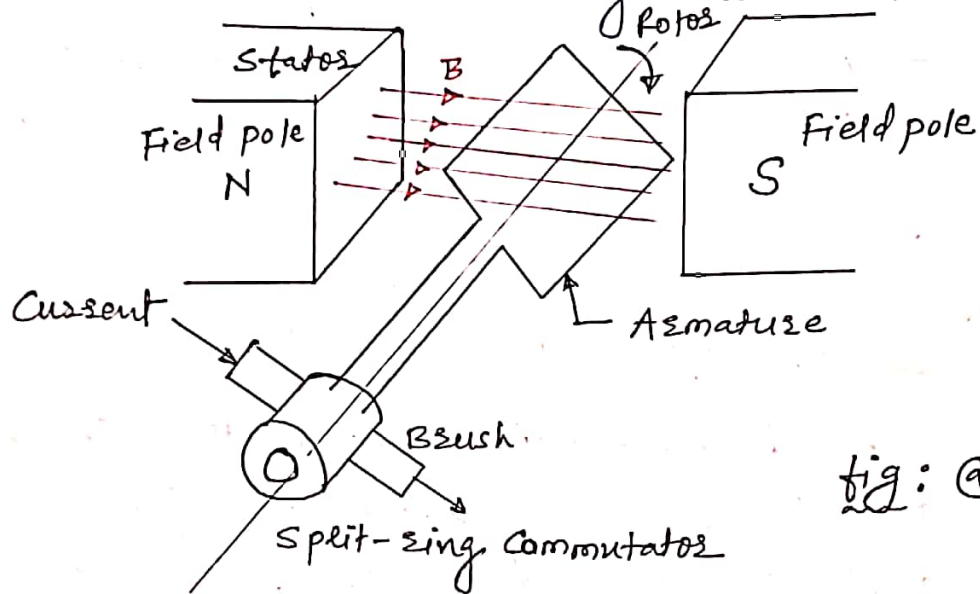


fig: (a)

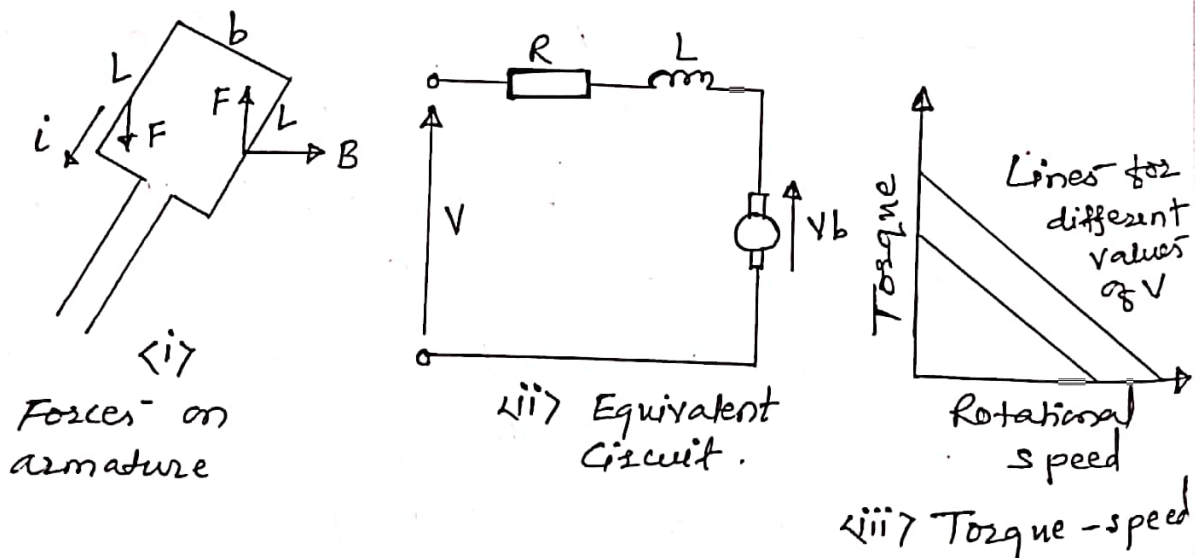


fig: (b)

- Consider a permanent magnet d.c. motor, the permanent magnet giving a constant value of flux density.
- For an armature conductor of length ' $L$ ' and carrying a current ' $i$ ', the force resulting from a magnetic flux density ' $B$ ' at right angles to conductor is  $BiL$  (fig: 6). With ' $N$ ' such conductors the force is  $NBiL$ . The force results in a torque ' $T$ ' about the axis of rotation, with ' $b$ ' being the breadth of the coil. Thus,

$$\text{Torque } T = NBbLi = k_t \cdot i \quad - (1)$$

Where,  $k_t$  is the Torque Constant

- Since an armature coil is rotating in a magnetic field, electromagnetic induction will occur and a back e.m.f will be induced.
- The back e.m.f ' $V_b$ ' is proportional to the rate at which the flux linked by the coil changes & hence, for a constant magnetic field, is proportional to the angular velocity ' $\omega$ ' of the rotation

$$\therefore \text{back e.m.f } V_b = k_v \cdot \omega \quad - (2)$$

Where,  $k_v$  is the back e.m.f constant.

- In the equivalent circuit as shown in (fig:ii), i.e. the armature coil being represented by a resistor 'R' in series with an inductance 'L' in series with a source of back e.m.f. If we neglect the inductance of the armature coil then the voltage providing the current 'i' through the resistance is applied voltage V minus the back emf, i.e.  $V - V_b$ . Hence

$$i = \frac{V}{R} = \frac{V - V_b}{R}$$

$$\therefore i = \frac{V - k_v \cdot \omega}{R} \quad - (3)$$

Thus; Torque T from eqn (1) is

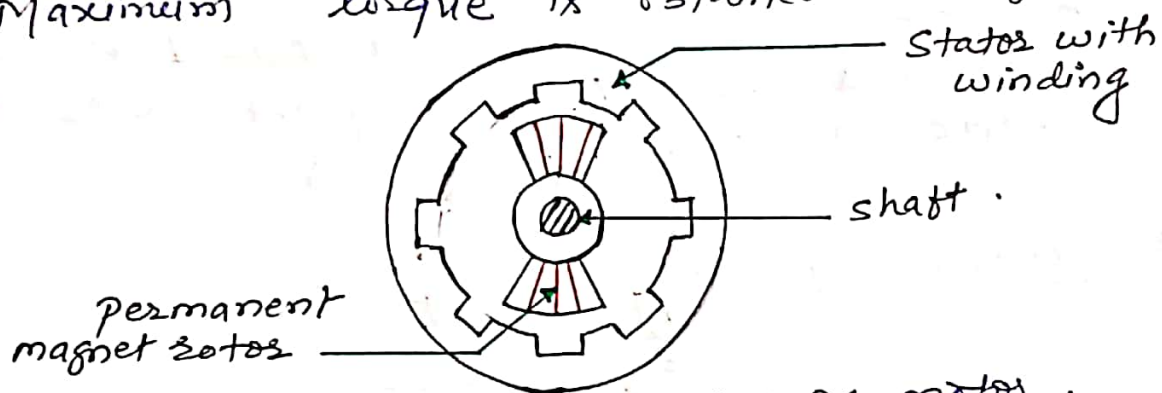
$$T = k_t \cdot i$$

$$\therefore T = \frac{k_t}{R} (V - k_v \cdot \omega) \quad - (4)$$

- Graphs of the torque against the rotational speed ' $\omega$ ' are a series of straight lines for different voltages.
- The starting torque, i.e. the torque when  $\omega = 0$ , is thus proportional to the applied voltage and the torque decreases with increase in speed.

## ⑥ Brushless - type D.C motor : [BLDC]

- BLDC consists of a stator coil and a permanent magnet rotor.
- The stator coil is fixed while, permanent magnet rotor rotates.
- BLDC stands for brushless d.c motor.
- The stator current is controlled by semi-conductor control, in such a way that maximum torque is obtained at a given speed.

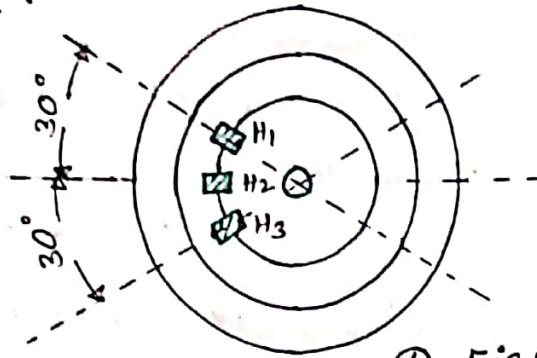


⑦ Fig: Construction of BLDC motor.

- Stator has 3-phase winding while rotor is in the form of permanent magnet.
- BLDC motor also consists of a rotor position sensors which produce electrical signals that indicate the current position of the rotor.
- BLDC stator winding is driven from an electronic drive which is 3-phase inverter.
- Speed of BLDC can be controlled by controlling its stator voltage i.e. by controlling the d.c input voltage of the inverter.

## Sensors:

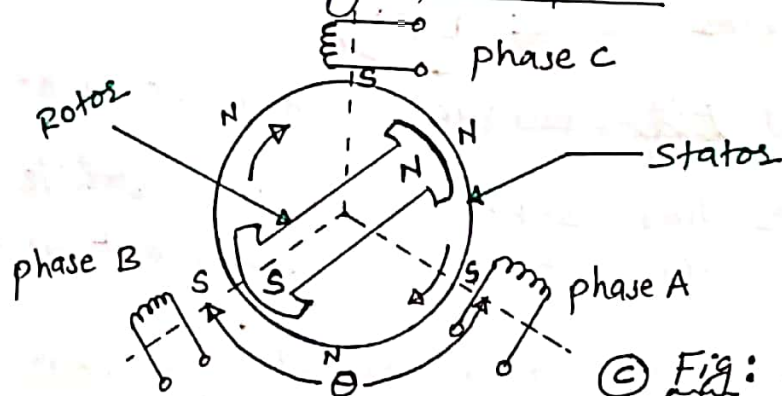
- The permanent magnet brushless d.c motors use Hall sensors with  $60^\circ$  electrical spacing.



⑥ Fig: Hall sensors

- These sensors produce a Logic "1" when exposed to N-type of the rotor and Logic "0" otherwise (S-type).

## #. Basic Operating principle:



⑦ Fig: Stator 3-phase winding A, B, C.

- Stator consists of 3-phase winding A, B, C as shown in fig.
- When phase A is energized, stator S pole repels rotor S pole & attracts rotor N-pole.
- When phase B is energized, stator S pole repels rotor S pole & attracts rotor N-pole.

→ Similarly, when phase 'c' is energized, stator S-pole repels rotor S-pole & attracts rotor N-pole.

→ In this way, the rotation of rotor takes place in clockwise direction.

→ Hence, clockwise Torque is produced.

→ The Magnitude of this torque is given by

$$T = k \phi_s \phi_r \sin \theta \quad - (1)$$

Where  $k$  = Constant

$\phi_s$  = stator field flux.

$\phi_r$  = rotor field flux.

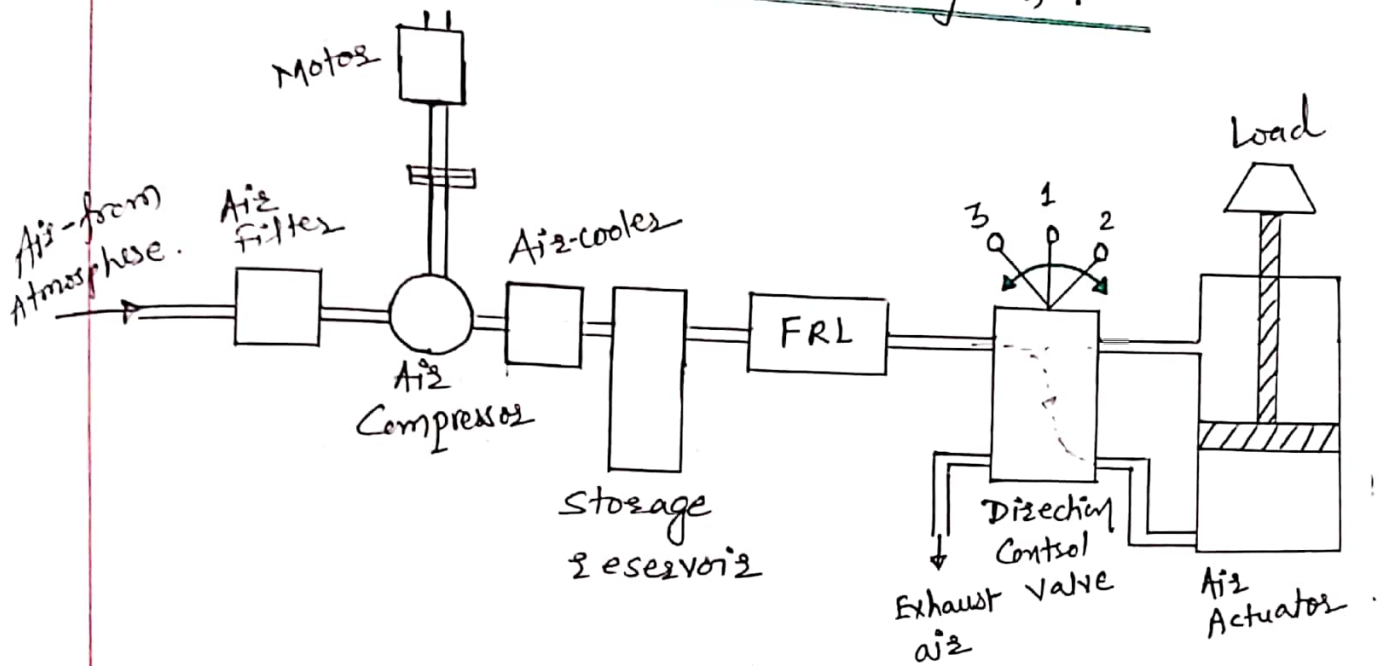
$\theta$  = torque angle.

→ The stator phase are energized in the sequence A, B, C, A so as to have clockwise rotation of the machine.

### Applications:

- a) Computer peripheral equipments.
- b) Instrumentation & control systems.
- c) Electric power steering
- d) Air Conditioners.
- e) Aerospace.

### 3. Components of Pneumatic System :



1 - off  
2 - forward  
3 - Return.

#### ① Air Filters:

— These are used to filter out the contaminants from the air.

#### ② Air Compressors:

— Compressed air is generated by using air compressors.

#### ③ Electric Motor:

— Transforms electrical energy into mechanical energy. It is used to drive the Compressor.

#### ④ Air Cooler:

— During compression operation, air temperature increases. Therefore coolers are used to reduce the temperature of the compressed air.

⑤. Reservoir :

— The Compressed air coming from the Compressor is stored in the air Reservoir.

⑥. FRL :

— Filter Regulator Lubricator. Before this air can be used, it needs to be cleaned and lubricated. That's where an FRL comes in.

⑦. Control Valves :

— Control valves are used to regulate, control and monitor of direction flow, Pressure etc.

⑧. Air Actuators :

— Cylinders and motors are used to obtain the required movements of mechanical elements of pneumatic system.

## 5. ACTUATORS :

- It is a device that causes a machine or other device to operate.

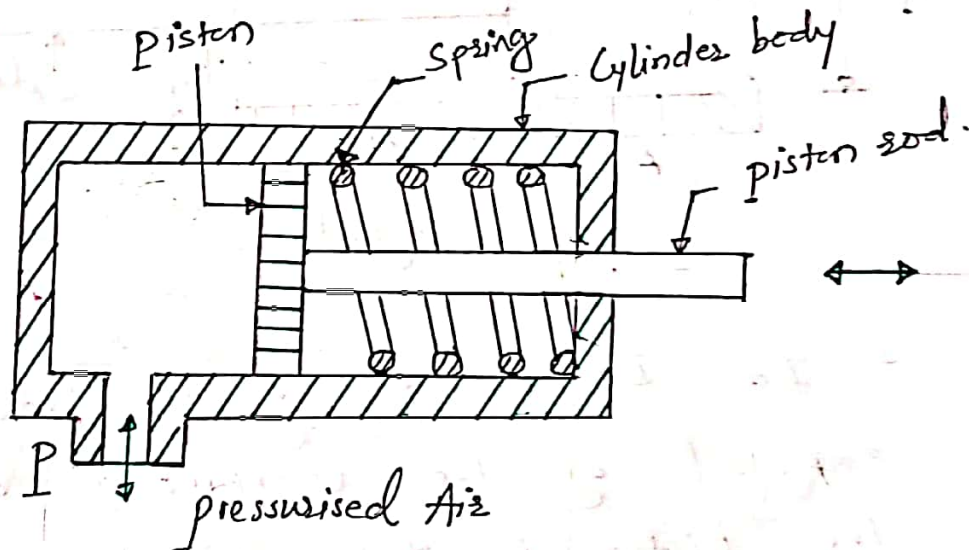
or

- An actuator is a component of a m/c that is responsible for moving and controlling a mechanism or system.

### A] Linear actuators:

- The hydraulic or pneumatic cylinder is an example of linear actuator.

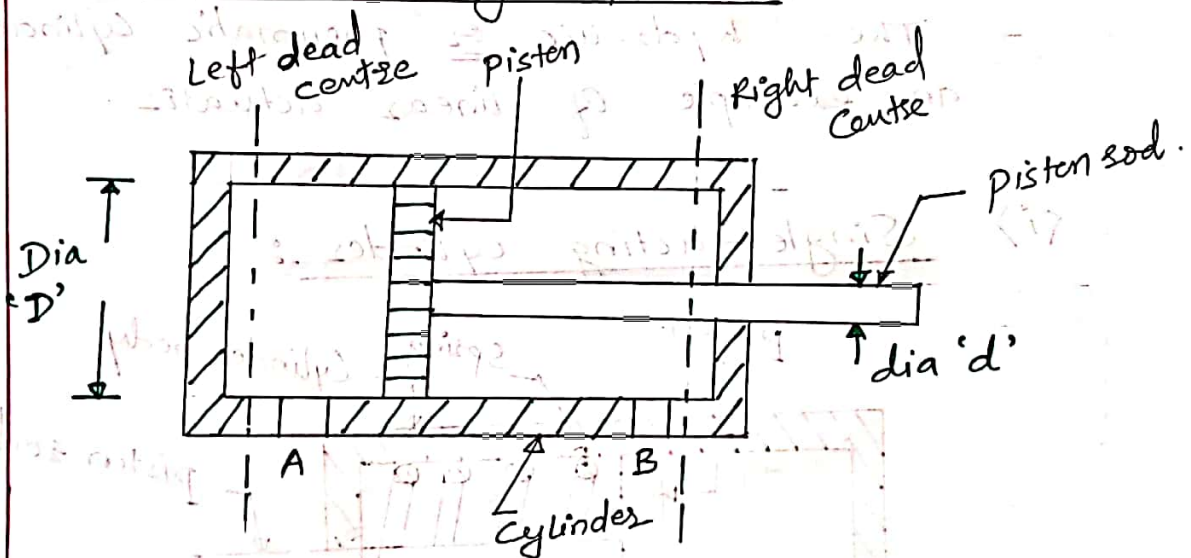
#### (i) Single-acting Cylinder :



- In a single acting cylinder, the pressurised air or compressed air is fed only in one side. Hence, this cylinder can produce work only in one direction. (right-dir<sup>n</sup>)
- The return movement of the piston is effected by a built in spring or by application of an external force.

- The spring is designed to return the piston to its initial position with sufficiently high speed.
- The movement of piston rod is used to do some useful work (e.g. pushing the load.)

### (iii) Double Acting Cylinder :

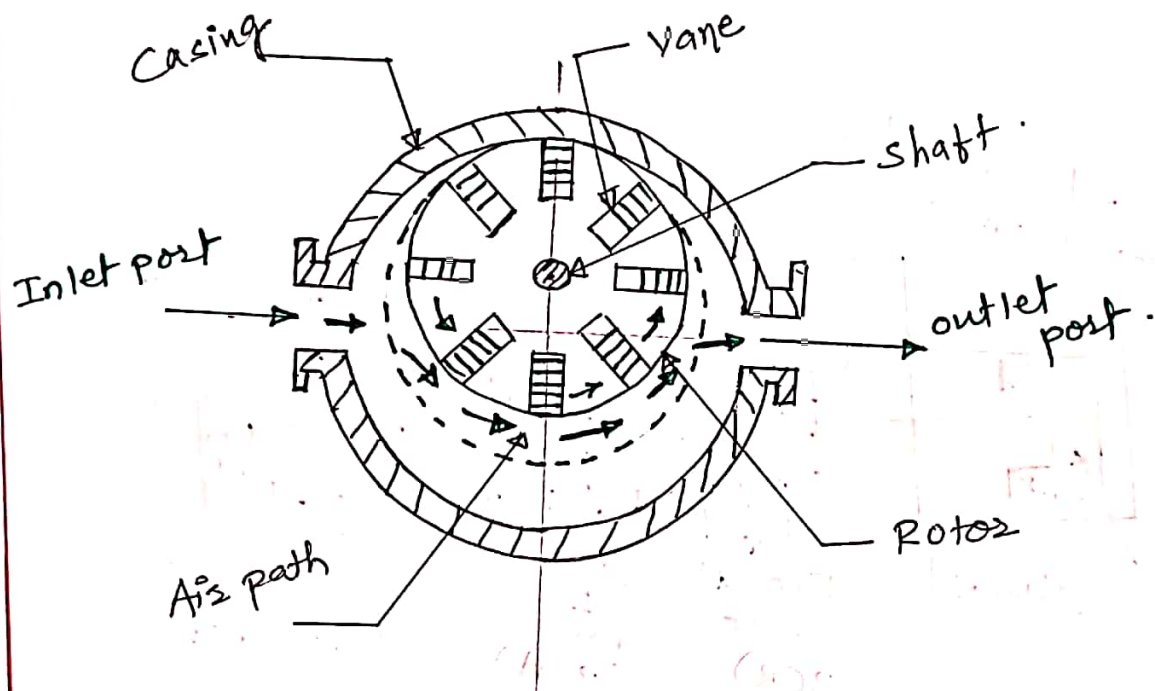


- In Double-Acting Cylinder, the force exerted by the compressed air moves the piston in two directions.
- A compressed air is passed through port 'A', then piston moves towards Right dead centre.
- When piston reaches (RDC) then air supply through port 'A' stops & compressed air is passed through port 'B'.

- When Compressed air is passed through port 'B', the piston moves towards left dead Centre.
- The movement of piston rod is used to do some useful mechanical work.
- In Double-acting Cylinder Spring is not used for return stroke, since pressure is applied from both sides i.e port 'A' & port 'B'.

## B] Rotary Actuators :

### (i) Vane motor :



— A vane motor consists of a rotor mounted on a shaft. Rotor consists of a vane as shown in a figure.

- When motor is started, the rotor is rotated and vacuum is created at the air path & air is sucked from the inlet port.
- The air is pressurised by the rotating vanes and high pressure air is delivered from the outlet port.
- The delivered high press. air is used to do some useful work.

## 6. Electro-pneumatic valves:

### a) Solenoid-operated spool valve:

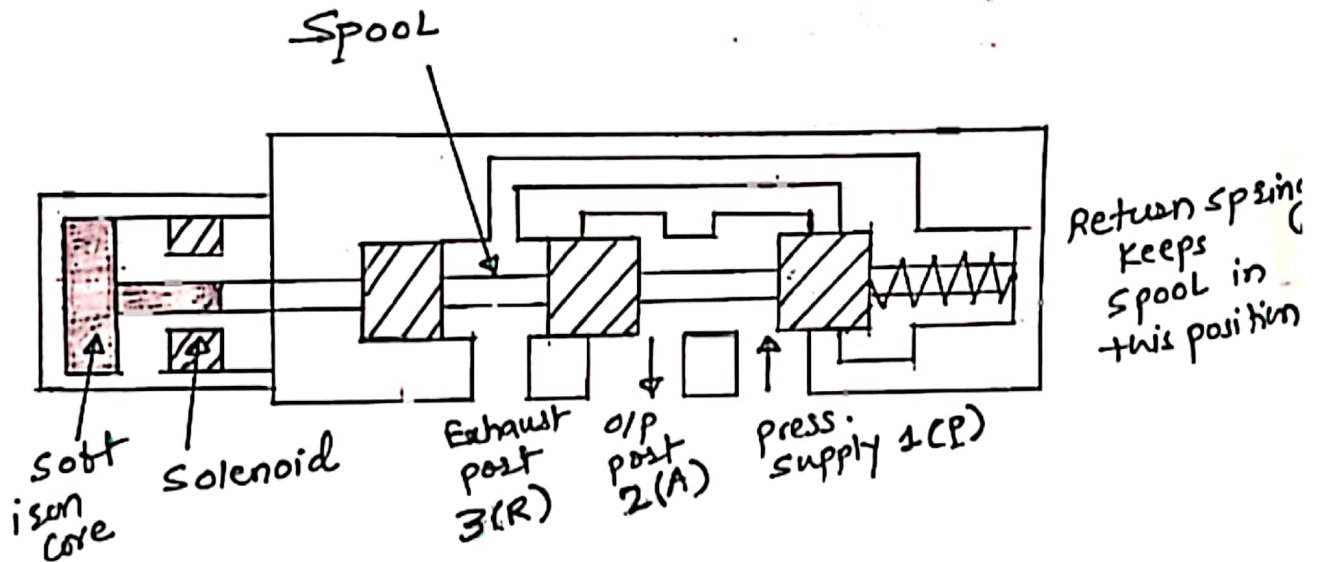


fig: (a)

Solenoid activated & spool moves.

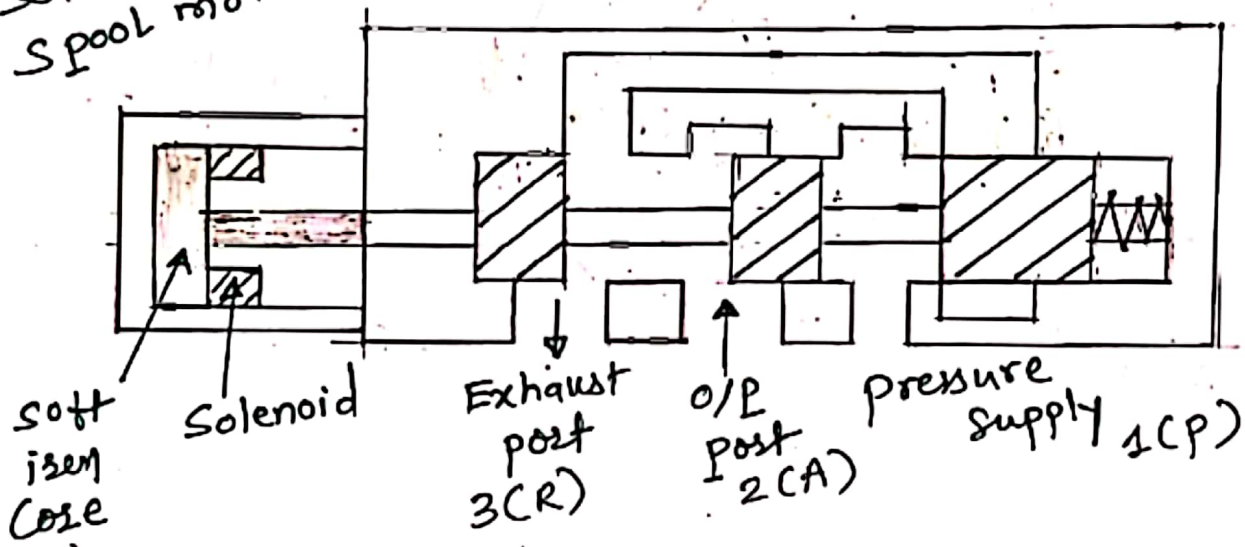


fig: (b)

⇒ i) Fig: shows a solenoid-operated spool valve & its symbol. The valve is actuated by a current passing through a solenoid & returned to its original position by a spring.

ii) Initially, 1(P) is connected to port 2(A) and Exhaust port 3(R) is closed as shown in fig: (a)

iii) When the solenoid is activated, the port 2(A) is connected to Exhaust port 3(R) and 1(P) is closed as shown in fig: (b)

iv) When the current through the solenoid ceases, the spring pushes the valve back to its initial position.

v) Thus, the movement of spring indicates the position of valve which is electrically operated.

# 1. Sequencing of pneumatic cylinders using Cascade method:

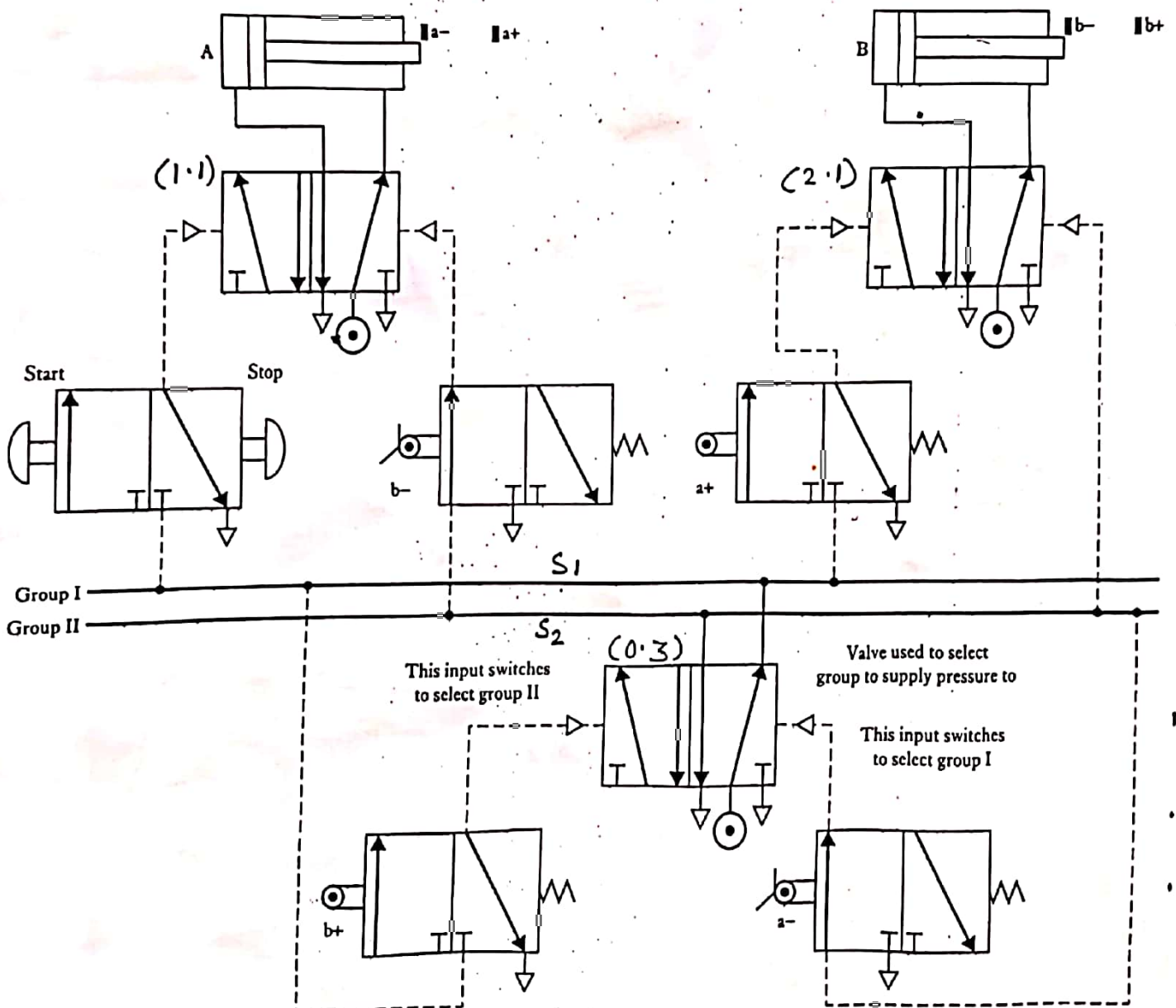
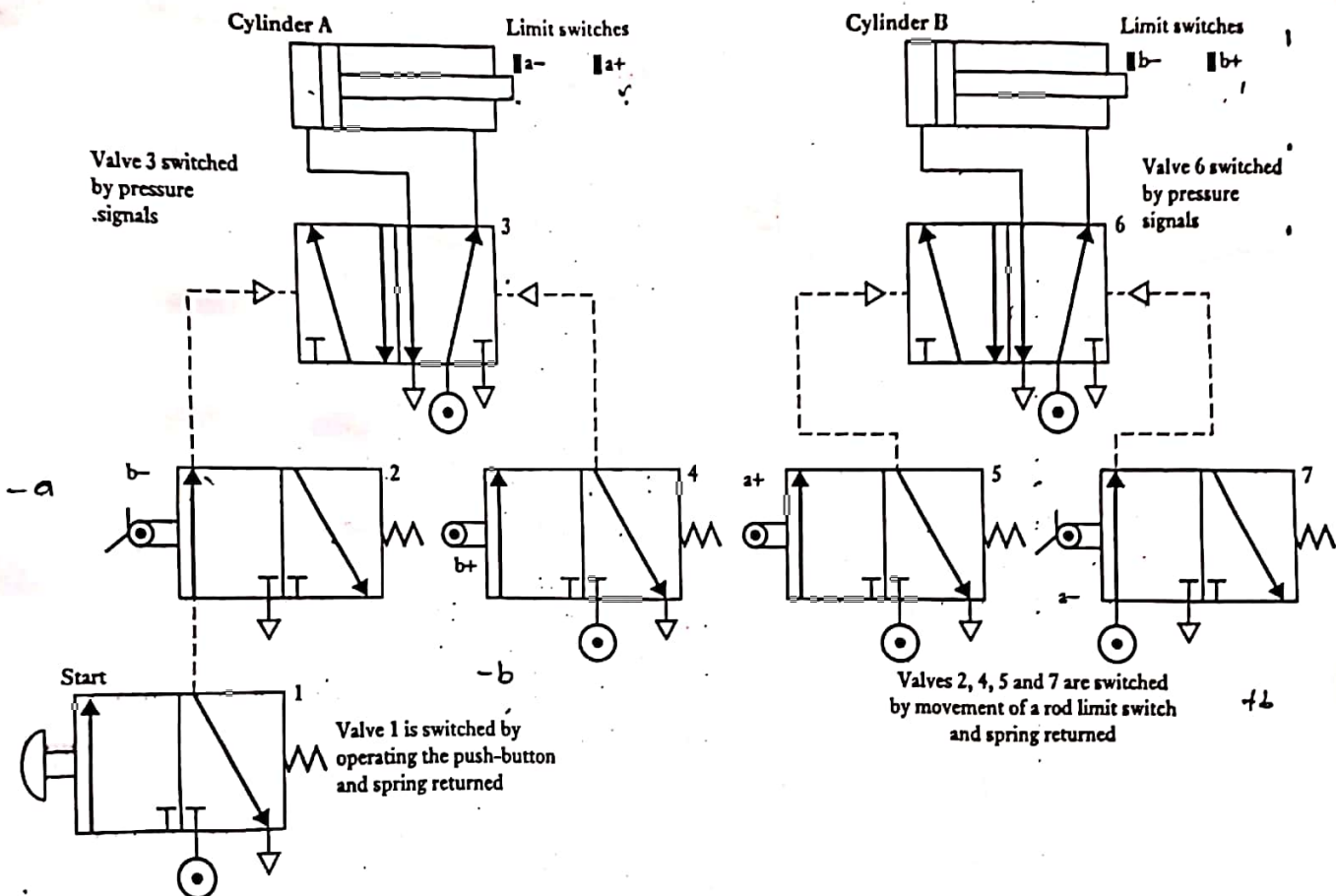


Figure 7.21 Cascade control used to give A+, B+, B-, A-

- The Cascade circuit diag. is shown in figure.
  - The groups are supplied (press.) by a 5/2 directional valve, called a group selector valve (0.3).
- i) Initially, the i/p switch (a-) is active since the cylinder A is retracted & the supply is fed from the line  $S_2$ .
  - ii) When start button is operated, the group selector valve (0.3) feeds line  $S_1$  with an air signal.
  - iii) The line  $S_1$  is directly connected to the valve (1.1) and cylinder A which extends valve and (a+) is now actuated & feeds signal from  $S_1$  to valve (2.1) which then extends cylinder B.
  - iv) When cylinder B extends, the limit switch (b+) is actuated. Due to this, the o/p signal is generated to reverse the switching of a group selector valve (0.3). The o/p signal switches the supply line from line ' $S_1$ ' to line ' $S_2$ '.
  - v) When line ' $S_2$ ' is energized or actuated, causes reversal of valve (2.1) & therefore cylinder B retracts. When cylinder B is fully retracted, the limit valve (b-) becomes active & reverses the valve (1.1) which retracts cylinder A to its initial position.

Sequencing of pneumatic cylinders-  
using shift register method:



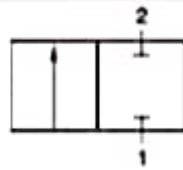
**Figure 7.20** Two-actuator sequential operation.

THE SEQUENCE OF OPERATION : ( $A^+$ ,  $B^+$ ,  $A^-$ ,  $B^-$ ) .

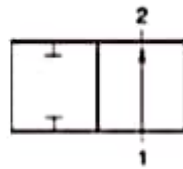
1. When push button is started , valve 1 is pressed. pressure is applied to valve 2 . Hence valve 3 is switched to apply pressure to cylinder A for extension .
2. Cylinder A extends (from  $a^-$  to  $a^+$ ) . This switches valve 5 and causes pressure to be applied to valve 6 and so apply pressure to cylinder B to cause its piston to extend .
3. Cylinder B extends (from  $b^-$  to  $b^+$ ) . This switches valve 4 and causes pressure to be applied to valve 3 and so applies pressure to cylinder A to start its piston to retracting .
4. Cylinder A retracts (from  $a^+$  to  $a^-$ ) . This switches valve 7 and causes pressure to be applied to valve 6 and so applies pressure to cylinder B to start its piston retracting .
5. Cylinder B retracts (from  $b^+$  to  $b^-$ ) . when cylinder B is fully retracted the cycle is completed .

## Symbols of the most common valves

**2/2-way valve**

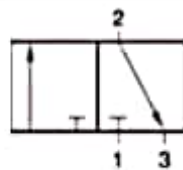


Normally closed

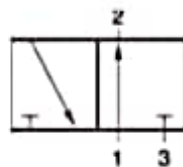


Normally open

**3/2-way valve**

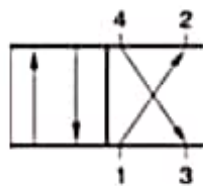


Normally closed

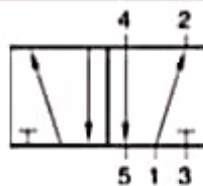


Normally open

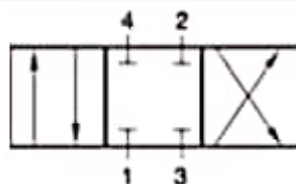
**4/2-way valve**



**5/2-way valve**

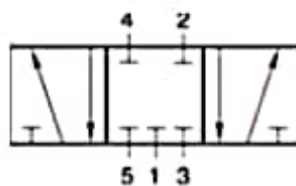


**4/3-way valve**



Center closed

**5/3-way valve**



Center closed

3.1

FRL Unit:

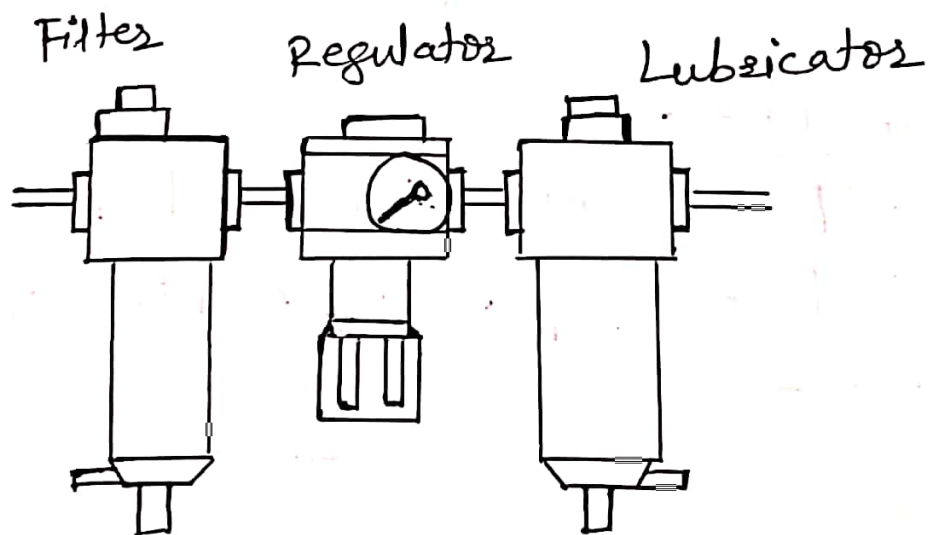


Fig: FRL unit.

## FRL Unit :

- It is the service unit used in pneumatic system which is the combination of three devices named as Filter, Regulator and Lubricator.

Filter (F) → To remove the micron & sub-micron particles present in the entering air of compressor. It is used to separate out contaminants like dust, dirt particles from the compressed air.

Regulator (R) → In pneumatic system the pressure of the compressed air may not be stable due to possibility of line fluctuation. Hence there is a need to maintain & regulate the air pressure. This function is performed by regulator.

Lubricator (L) → Sliding components like spool, a pneumatic cylinder have sliding motion between parts. It may cause friction & wear & tear at mating parts. To reduce friction, lubricating oil particles are added in the compressed air with the help of lubricator.

#### 4. Directional Control Valves :

- a) Spool valve
- b) Poppet valve

##### a) SPOOL VALVE :

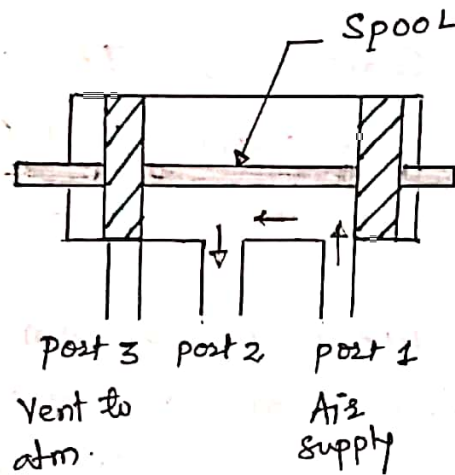


fig: (a)

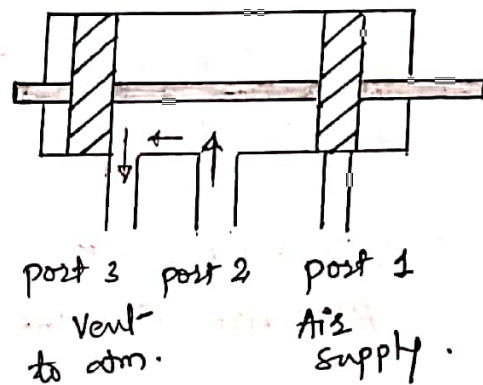
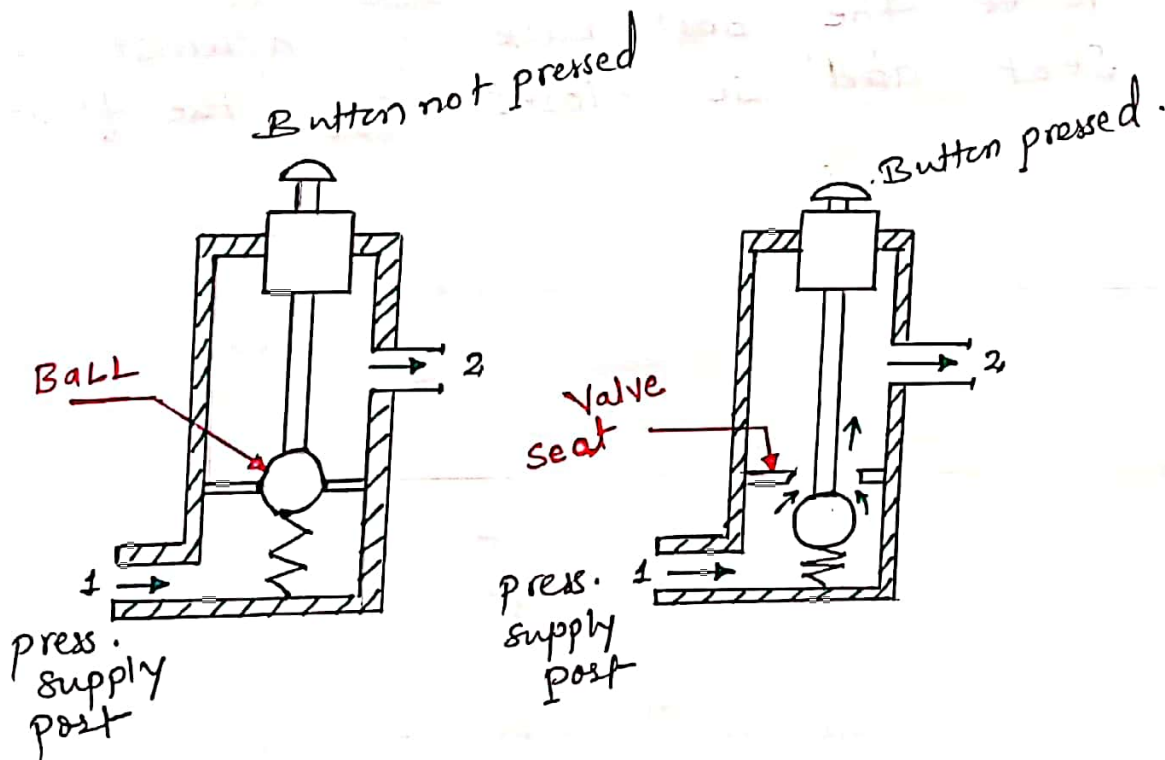


fig: (b)

- A Common type of directional control valve is the spool valve. A spool moves horizontally within the valve body to control the flow.
- In fig: (a) the air supply is connected to port (1) and port (3) is closed. Thus the device connected to port (2) can be pressurised.
- When the spool is moved towards left, as shown in fig: (b) the air supply is cut-off & port (2) is connected to port (3).

- port (3) is vent to atmosphere & so the air pressure in the system attached to port (2) is vented.
- Thus the movement of the spool has allowed the air firstly to flow into the system & then be reversed & flow out of the system.

## b) POPPET VALVE :



- Another common form of directional control valve is the poppet valve.
- This valve is normally in the closed condition, there being no connection between port 1 to which the pressure supply is connected & port 2 to which the system is connected.

- In poppet valves, balls, discs or cones are used in conjunction with valve seats to control the flow.
- In the figure a ball is shown. When the push-button is depressed, the ball is pushed out of its seat & flow occurs as a result of port ① being connected to port ②.
- When the button is released, the spring forces the ball back up against its seat and so closes off the flow.